



Send a Message

Quick facilitation notes & answer key

Goal

Students meet BROADCAST and RECEIVE: a script can send a message ('broadcast go') that starts another script ('when I receive go'). The message is itself an event — Pixel runs only because Bit broadcast 'go'. This is how separate scripts coordinate, and it is part of writing/executing code with events in Grade-2 C3.1.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: have one student be Bit and one be Pixel; Bit moves, then says 'go!' out loud — Pixel may only move after hearing 'go'.

Run it (about 20 min)

1. I DO: read Bit's script — 'green flag starts it, 0 forward 2 → 2, then it broadcasts go.'
2. WE DO: point to Pixel's hat — 'when I receive go'. 'Pixel was waiting; now it hears go and runs: 5 back 2 → 3.'
3. YOU DO: students find both landings and answer what starts Pixel (the message 'go'), and that with no message Pixel would not move.
4. Connect: 'A broadcast is a message; a receive block waits for it. The message is the event.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 2 → 2, then broadcasts 'go'. Bit lands on 2.

Ex 2 — Pixel receives 'go' and runs: 5 → back 2 → 3. Pixel lands on 3.

Ex 3 — Pixel is started by the message 'when I receive go'. If Bit never sent 'go', Pixel would NOT move (it stays home at 5).

Event note: the broadcast happens at the END of Bit's script, so Pixel starts after Bit — but it is the MESSAGE, not the green flag, that triggers Pixel.

Watch for & differentiate

Common error: thinking Pixel also starts on the green flag — Pixel has no green-flag hat; only the message 'go' starts it.

Common error: ignoring the broadcast block at the bottom of Bit's script — that block is what sends the message.

Simplify: act it out — Pixel literally cannot move until it hears 'go'.

Extend: ask what would happen if Pixel's hat said 'when I receive stop' but Bit broadcasts 'go' (the messages don't match → Pixel never starts — preview of debugging).

Success indicator: the child lands Bit on 2 and Pixel on 3 and explains the message 'go' is what starts Pixel.